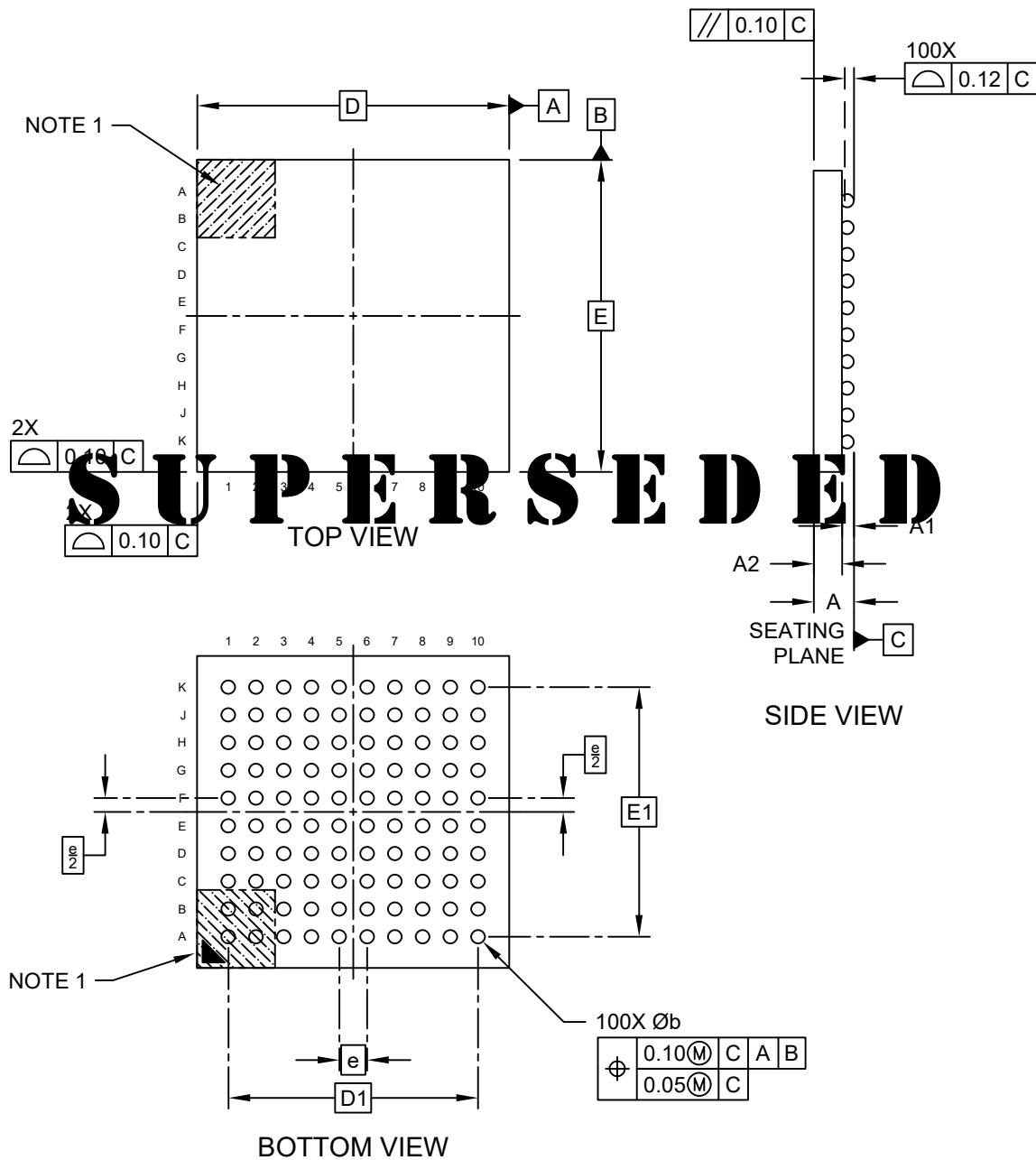


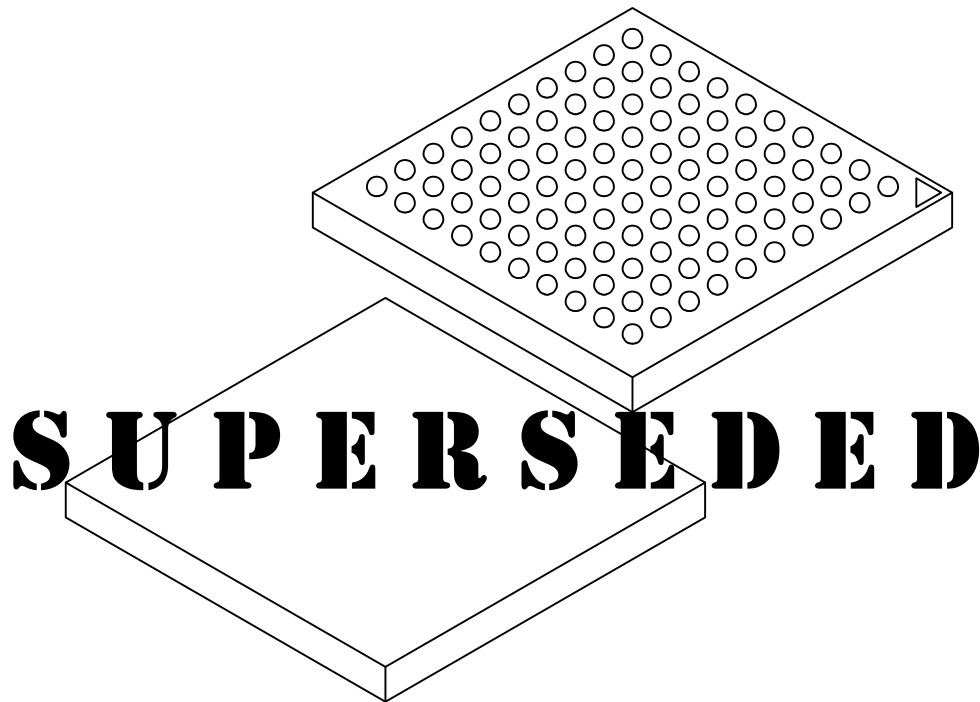
100-Ball Ceramic Ball Grid Array Package (A2B) - 9x9 mm Body [CBGA] **Atmel Legacy Global Package Code CPR**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



100-Ball Ceramic Ball Grid Array Package (A2B) - 9x9 mm Body [CBGA]
Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



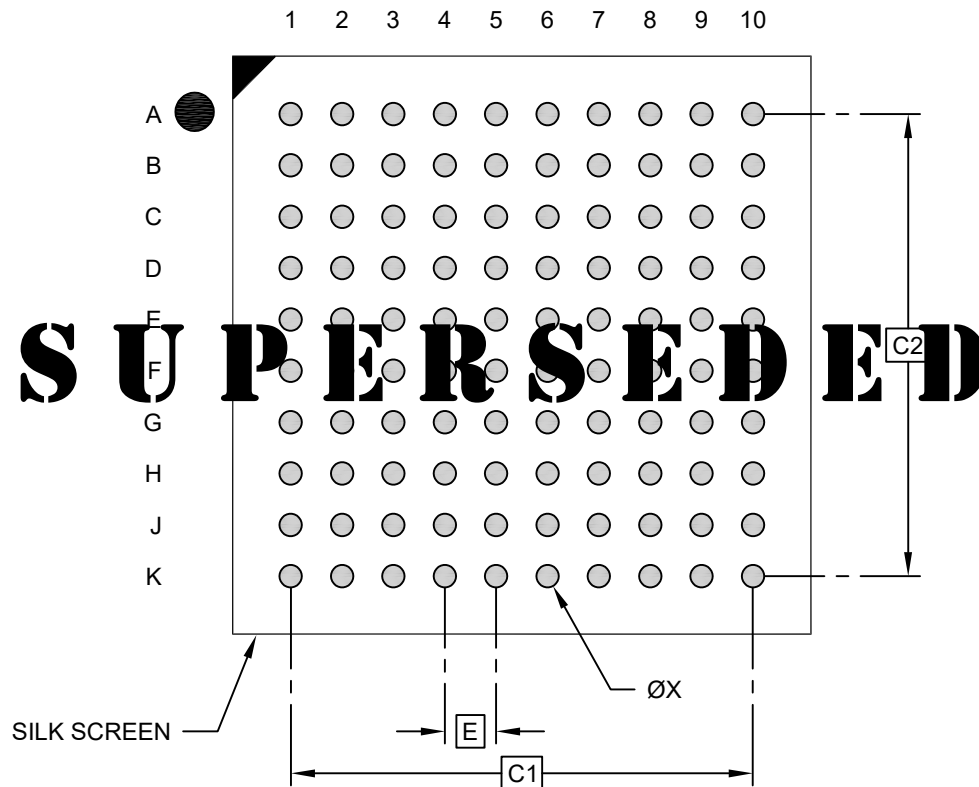
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.80 BSC		
Overall Height	A	1.10	-	1.20
Ball Height	A1	0.20	-	-
Molded Package Height	A2	0.65	-	-
Overall Length	D	9.00 BSC		
Overall Pitch	D1	7.20 BSC		
Overall Width	E	9.00 BSC		
Overall Pitch	E1	7.20 BSC		
Terminal Diameter	b	0.40	0.45	0.50

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

100-Ball Ceramic Ball Grid Array Package (A2B) - 9x9 mm Body [CBGA] Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.58 BSC		
Overall Pitch	C1	7.20 BSC		
Contact Pad Spacing	C2	7.20 BSC		
Contact Pad Diameter (X100)	X1			0.35

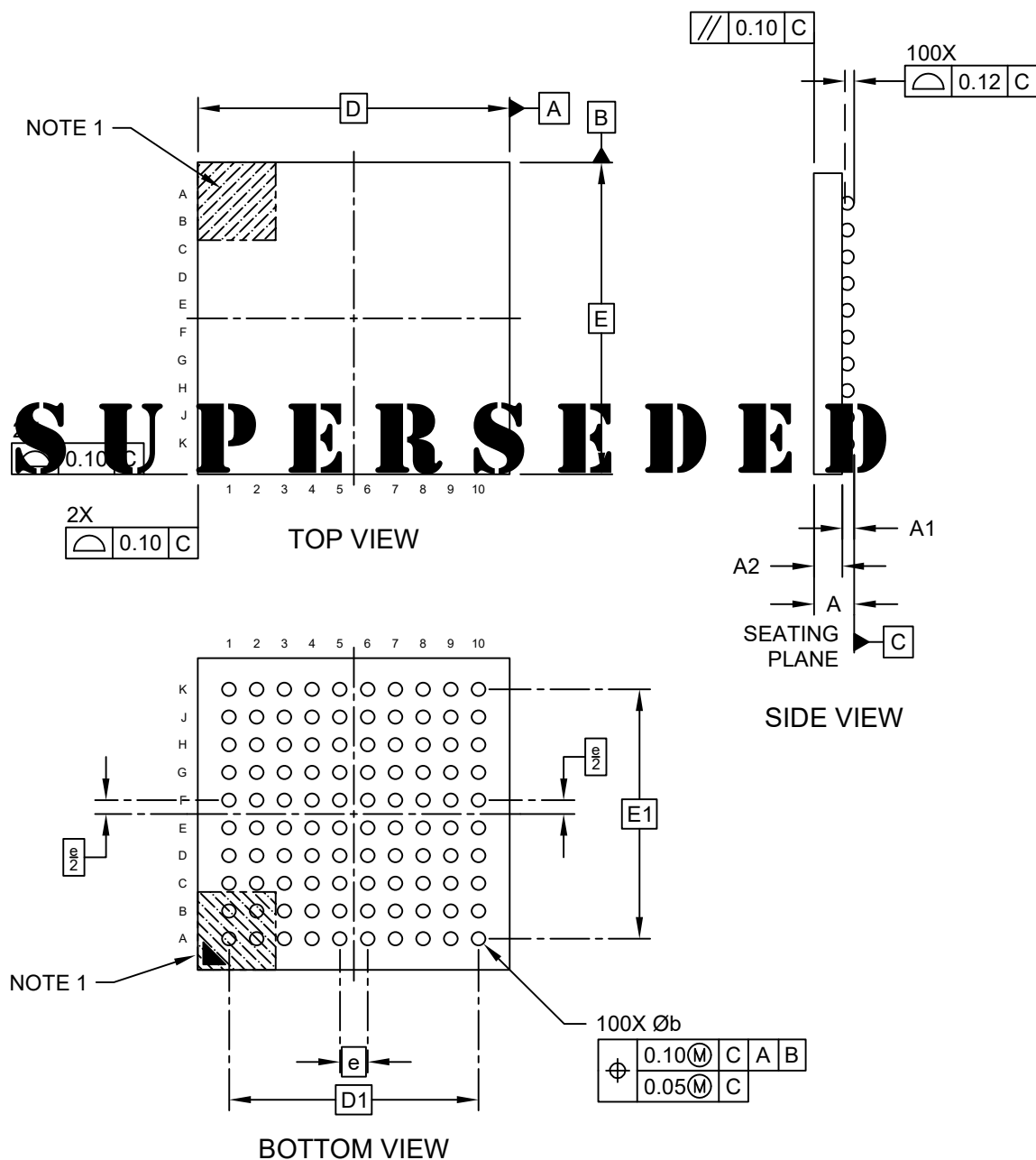
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23111-A2B Rev C

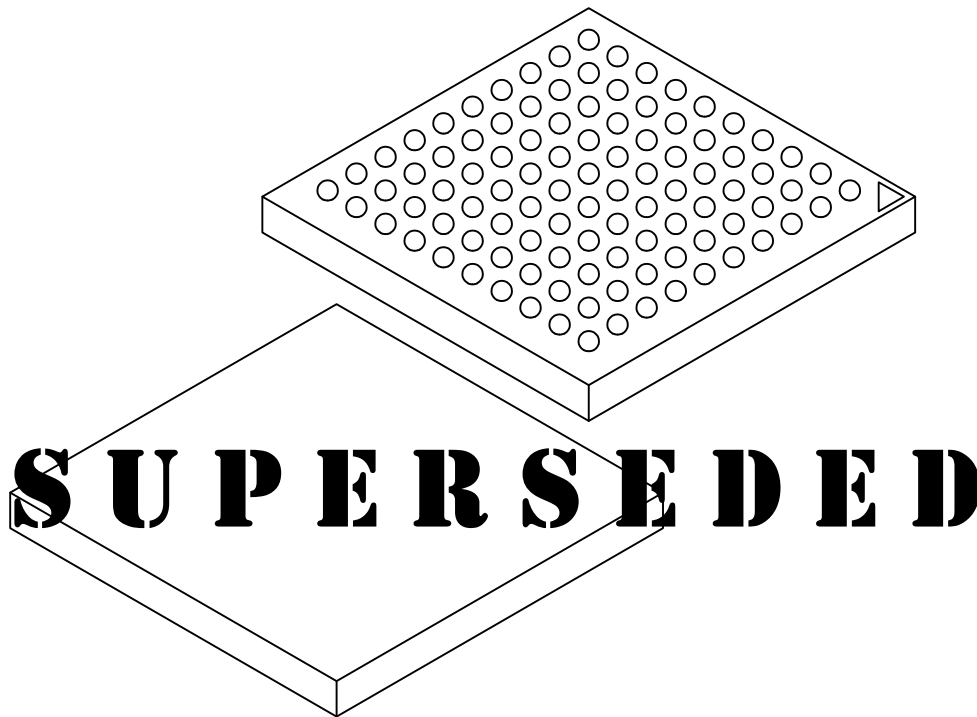
100-Ball Ceramic Ball Grid Array Package (A3B) - 9x9 mm Body [CBGA] Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



100-Ball Ceramic Ball Grid Array Package (A3B) - 9x9 mm Body [CBGA]
Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



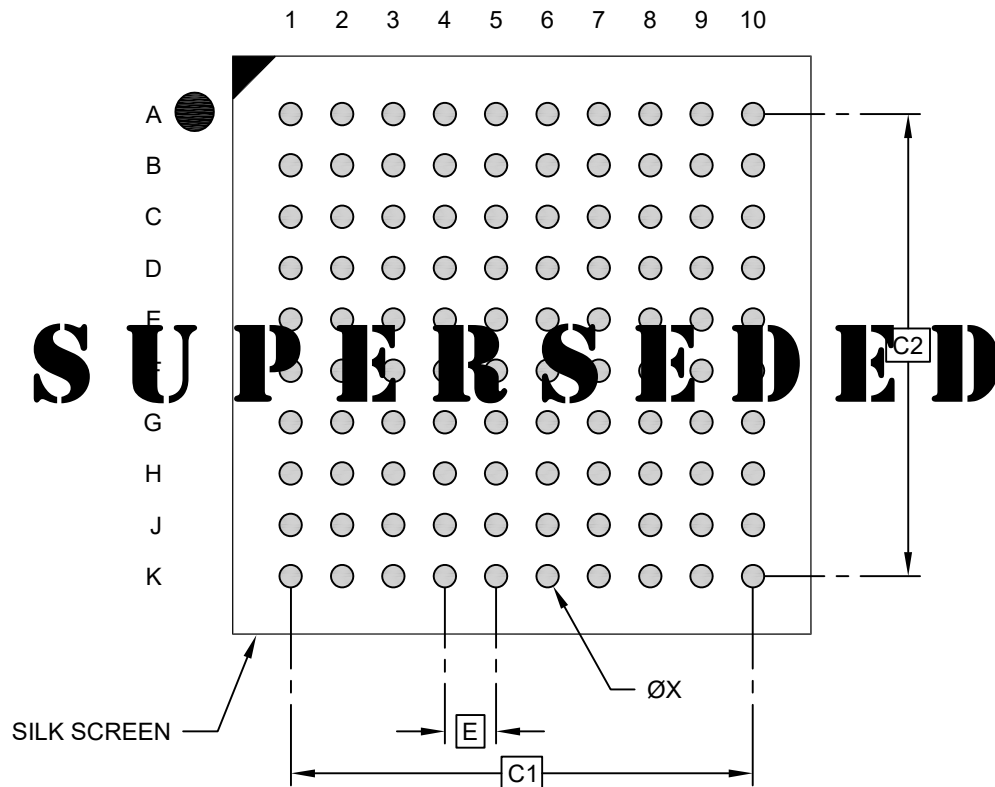
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.80 BSC		
Overall Height	A	1.10	-	1.20
Ball Height	A1	0.20	-	-
Molded Package Height	A2	0.65	-	-
Overall Length	D	9.00 BSC		
Overall Pitch	D1	7.20 BSC		
Overall Width	E	9.00 BSC		
Overall Pitch	E1	7.20 BSC		
Terminal Diameter	b	0.40	0.45	0.50

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

100-Ball Ceramic Ball Grid Array Package (A3B) - 9x9 mm Body [CBGA] Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.58 BSC		
Overall Pitch	C1	7.20 BSC		
Contact Pad Spacing	C2	7.20 BSC		
Contact Pad Diameter (X100)	X1			0.35

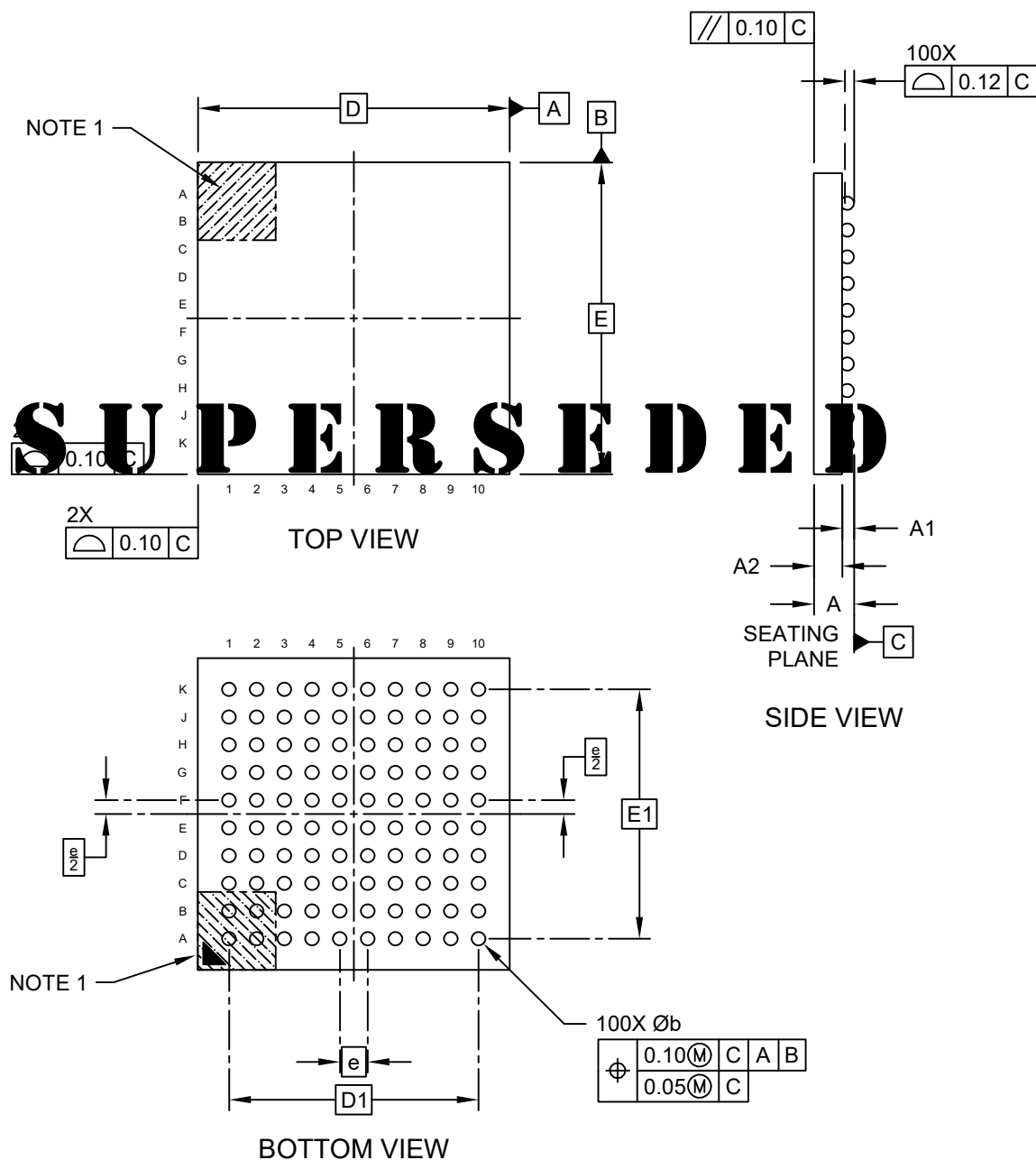
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23111-A3B Rev C

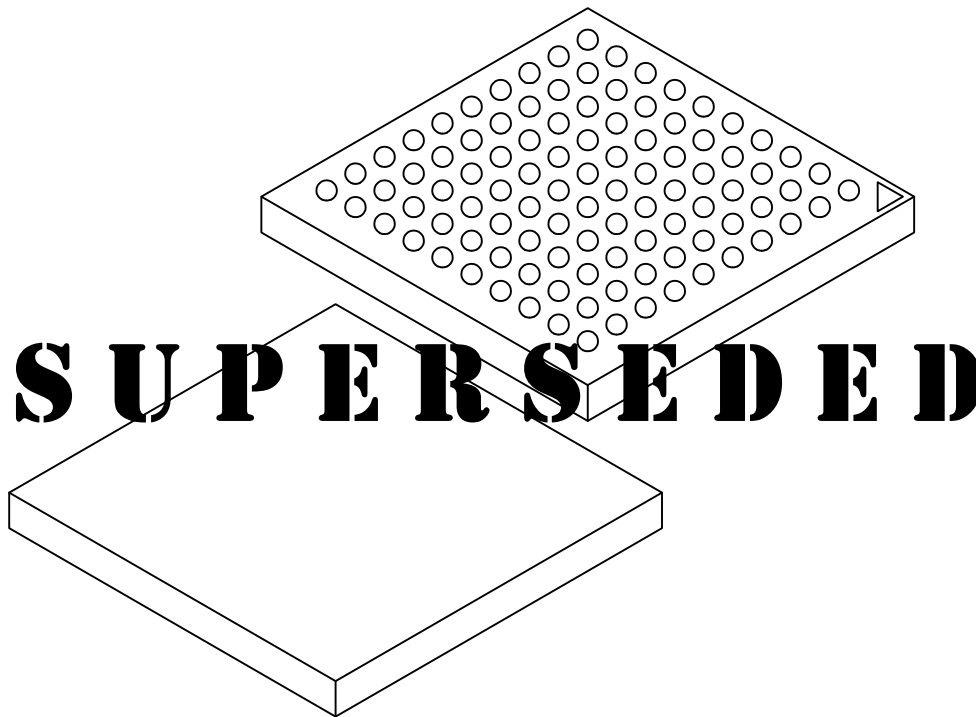
100-Ball Ceramic Ball Grid Array Package (CQB) - 9x9 mm Body [CBGA] **Atmel Legacy Global Package Code CPR**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



100-Ball Ceramic Ball Grid Array Package (CQB) - 9x9 mm Body [CBGA]
Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



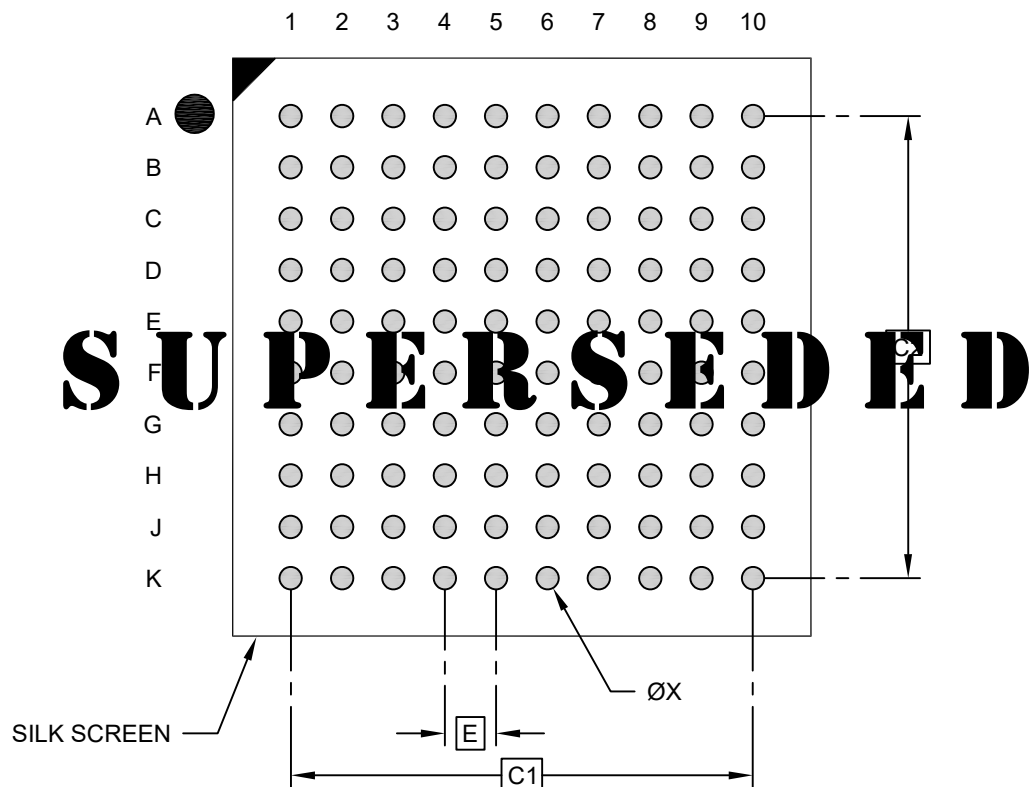
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.80 BSC		
Overall Height	A	1.10	-	1.20
Ball Height	A1	0.20	-	-
Molded Package Height	A2	0.65	-	-
Overall Length	D	9.00 BSC		
Overall Pitch	D1	7.20 BSC		
Overall Width	E	9.00 BSC		
Overall Pitch	E1	7.20 BSC		
Terminal Diameter	b	0.40	0.45	0.50

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

100-Ball Ceramic Ball Grid Array Package (CQB) - 9x9 mm Body [CBGA] Atmel Legacy Global Package Code CPR

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.58 BSC		
Overall Pitch	C1	7.20 BSC		
Contact Pad Spacing	C2	7.20 BSC		
Contact Pad Diameter (X100)	X1			0.35

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23111-CQB Rev A